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Java SE 27 & JDK 27

Interface Channel

All Superinterfaces:

AutoCloseable, Closeable

All Known Subinterfaces:

AsynchronousByteChannel, AsynchronousChannel, ByteChannel, GatheringByteChannel, InterruptibleChannel, MulticastChannel, NetworkChannel, ReadableByteChannel, ScatteringByteChannel, SeekableByteChannel, WritableByteChannel

All Known Implementing Classes:

AbstractInterruptibleChannel, AbstractSelectableChannel, AsynchronousFileChannel, AsynchronousServerSocketChannel, AsynchronousSocketChannel, DatagramChannel, FileChannel, Pipe.SinkChannel, Pipe.SourceChannel, SctpChannel, SctpMultiChannel, SctpServerChannel, SelectableChannel, ServerSocketChannel, SocketChannel

---

```
public interface Channel
extends Closeable
```

A nexus for I/O operations.

A channel represents an open connection to an entity such as a hardware device, a file, a network socket, or a program component that is capable of performing
one or more distinct I/O operations, for example reading or writing.

A channel is either open or closed. A channel is open upon creation, and once closed it remains closed. Once a channel is closed, any attempt to invoke an I/O operation upon it will cause a ClosedChannelException to be thrown. Whether or not a channel is open may be tested by invoking its isOpen method.

Channels are, in general, intended to be safe for multithreaded access as described in the specifications of the interfaces and classes that extend and implement this interface.

Since:

1.4

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description

void	**close**()	Closes this channel.
boolean	**isOpen**()	Tells whether or not this channel is open.

Method Details

isOpen

```
boolean isOpen()
```

Tells whether or not this channel is open.

Returns:

`true` if, and only if, this channel is open

close

```
void close()
    throws IOException
```

Closes this channel.

After a channel is closed, any further attempt to invoke I/O operations upon it will cause a `ClosedChannelException` to be thrown.

If this channel is already closed then invoking this method has no effect.

This method may be invoked at any time. If some other thread has already invoked it, however, then another invocation will block until the first invocation is complete, after which it will return without effect.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Throws:

`IOException` - If an I/O error occurs

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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